Linear Maps and a Uniform Bound

1 Norms and linear maps

This note studies linear maps between finite-dimensional real vector spaces. Throughout, $m, n \geq 1$ and $A \in \mathbb{R}^{m \times n}$. 中文说明：本讲义同时包含中英文文字。

Definition 1.1 (Euclidean norm)**.** *For $x = (x_1, \ldots, x_n) \in \mathbb{R}^n$, its Euclidean norm is*

$$\|x\|_2 = \left(\sum_{j=1}^{n} x_j^2 \right)^{1/2}. \tag{1}$$

The linear map associated with A is $T_A(x) = Ax$. Every statement below uses the norm in Definition 1.1.

$$(Ax)_i = \sum_{j=1}^{n} a_{ij} x_j, \qquad 1 \leq i \leq m. \tag{2}$$

No assumption of invertibility is imposed. In particular, a zero matrix is allowed.[1]

[1] All vector spaces in this note are over the real numbers.

2 A uniform bound

Theorem 2.1. *For every $x \in \mathbb{R}^n$, one has*

$$\|Ax\|_2 \le C\|x\|_2, \qquad C = \left(\sum_{i=1}^{m} \sum_{j=1}^{n} a_{ij}^2 \right)^{1/2}. \tag{3}$$

证明. By the Cauchy–Schwarz inequality, for each i,

$$\left(\sum_{j=1}^{n} a_{ij} x_j \right)^2 \le \left(\sum_{j=1}^{n} a_{ij}^2 \right) \left(\sum_{j=1}^{n} x_j^2 \right).$$

Summing over i and taking nonnegative square roots proves (3). This argument also applies when $C = 0$. □

Theorem 2.1 implies continuity: if $x_k \to x$, then $\|Ax_k - Ax\|_2 \le C\|x_k - x\|_2 \to 0$. The estimate is sufficient for continuity; it need not be the smallest possible bound.

3 Geometry and an exercise

For $A = \mathrm{diag}(2, 1)$, the unit circle is mapped to the ellipse in Figure 1. The horizontal radius doubles and the vertical radius is unchanged.

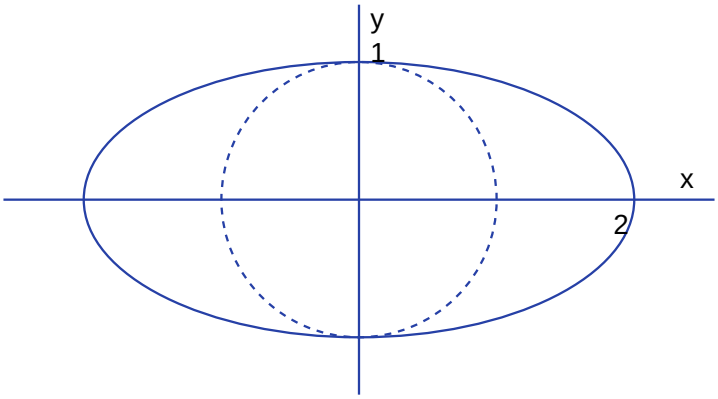

图 1: The image of the unit circle under $A = \mathrm{diag}(2, 1)$.

x	Ax	$\|Ax\|_2$
$(1, 0)$	$(2, 0)$	2
$(0, 1)$	$(0, 1)$	1

Exercise 3.1. *Let $A = \mathrm{diag}(2, 1)$. Find the smallest $K \geq 0$ such that $\|Ax\|_2 \leq K\|x\|_2$ for every $x \in \mathbb{R}^2$. Compare your answer with C in (3).*

4 Review and references

The order of quantifiers matters: one constant C works for every vector x. The claim does not require A to be invertible and does not assert that C is optimal.

1. Identify the variables and indices in (2).
2. Explain why taking square roots in the proof preserves the inequality.
3. Check that Figure 1 agrees with the table.

For the source of this self-contained note, see [1]. The bibliography identifies the local source; no external mathematical citation is needed for the proof given here.

References

[1] *Linear Maps and a Uniform Bound*. Local original teaching note, CC0 1.0. 2026.